CP–7886

Sub. Code
11T

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

தமிழ்ச் செம்மொழியும் தமிழர்களின் பன்முகத் திறனும்

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

பகுதி அ (10 × 2 = 20)

எல்லா வினாக்களுக்கும் விடை தருக.

1. தமிழின் சிறப்புக்கள் இரண்டினைத் தருக.

2. உலகச் செம்மொழிகள் எவை?

3. ஆடை குறித்த பெயர்களைத் தருக.

4. ஒற்றை ஆடை என்பதனை விளக்குக.

5. ‘கிண்கிணி’ விளக்கம் தருக.

6. ஆண்கள் அணியும் அணிகலன்களின் பெயர் தருக.

7. ‘யாழ்’ என்னும் இசைக்கருவியின் தன்மையை விவரி.

8. சிற்பங்கள் பற்றி குறிப்பு வரைக.

9. பழந்தமிழ் கடற்கலங்களின் பெயர்களைத் தருக.

10. ‘கணி’ என்பதன் விளக்கம் கூறுக.

பகுதி ஆ $(5 \times 5 = 25)$

எல்லா வினாக்களுக்கும் விடை தருக.

11. (அ) இந்தியச் செம்மொழிகள் பற்றி குறிப்பு வரைக.

(அல்லது)

(ஆ) தமிழ்ச் செம்மொழி நூல்கள் பற்றி குறிப்பு வரைக.

12. (அ) ஆடை வகைகளை விளக்குக.

(அல்லது)

(ஆ) ஆடை வேலைப்பாடுகள் பற்றி எழுதுக.

13. (அ) அணி வகைகளை விளக்குக.

(அல்லது)

(ஆ) கழல் வகை அணிகளின் சிறப்புக்களைத் தருக.

14. (அ) ஓவியக்கலையின் தோற்றம் குறித்து உரைக்க.

(அல்லது)

(ஆ) மாதவி ஆடிய நடனம் பற்றி விவரிக்க.

15. (அ) பழந்தமிழரின் கல்வி பற்றிய செய்திகளைத் தருக.

(அல்லது)

(ஆ) சங்ககால மருத்துவம் பற்றி நீவிர் அறிவன யாவை?

பகுதி இ (3 × 10 = 30)

எல்லா வினாக்களுக்கும் விடை எழுதுக.

16. (அ) செம்மொழிக்கான தகுதிகளை வரையறை செய்க.

(அல்லது)

(ஆ) பழங்கால ஆடை பற்றிய செய்திகளைத் தொகுத்துரைக்க.

17. (அ) நூலாடை, பட்டாடை பற்றிய செய்திகளைச் சான்றுடன் விவரி.

(அல்லது)

(ஆ) பழந்தமிழகப் பெண்கள் அணிந்த அணிகலன்களாக நீவிர் அறிந்தவற்றைத் தொகுத்துரைத்திடுக.

18. (அ) சங்ககால கலைகள் பற்றி விவரிக்க.

(அல்லது)

(ஆ) பழந்தமிழர்களின் பல்துறை மேலாண்மை பற்றி புலப்படுத்துக.

CP–7887

Sub. Code
11A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Part I – ARABIC-BASIC GRAMMAR

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the names of three Arabic short vowels.

2. Write the three weak or irregular Arabic letters.

3. Give any two examples for the prepositions.

4. Mention any two examples for real feminine nouns.

5. Write any two examples for the Dual feminine.

6. Write any two examples for relative pronoun.

7. Write the Arabic equivalents for the following
   (a) Earth
   (b) Heavens
   (c) Paradise
   (d) Prayer.

8. Mention the two varieties of personal pronouns.

9. Write any two examples for Imperative.

10. Write any two examples for the passive voice.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Define Doubled consonants and absence of vowel with examples.

Or

(b) Define "Definite nouns of Arabic" with examples.

12. (a) Define "The Unreal or Supposed feminine" with examples.

Or

(b) Define "The natural or real feminine" with examples.

13. (a) Define "The past tense verbs" with examples.

Or

(b) Define "The Demonstrative Pronouns" with examples.

14. (a) Write any five examples for the negative command verbs in Arabic with their English equivalents.

Or

(b) Define "The Possessive" with examples.

15. (a) Define 'The transitive verbs" with examples.

Or

(b) Define Kaana and its categories with examples.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) The Arabic Alphabet and their changing shapes.

Or

(b) The adjective and the noun qualified.

17. (a) The Nominal sentence.

Or

(b) The Verbal sentence.

18. (a) Inna and its categories.

Or

(b) The Arabic numerals (Cardinals and Ordinals).

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

PROSE AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is education compared to?

2. Whose view on courage is quoted by Indira Gandhi?

3. Who said "Life is a bundle of Habits"?

4. Who is regarded as a gorilla according to the teacher in "Crime and Punishment"?

5. Who is Margaret Atwood?

6. What are the three visions of Mohammad Ali Jinnah on patriotism?

7. Identify the sentence type:

   (a) Taj Mahal is a symbol of Indian construction.

   (b) When will you return?

8. What is meant by noun? Give an example.

9. What is Indefinite Article?

10. Fill in the blanks with correct preposition :

(a) I jumped ———————— the river.

(b) The cat is ———————— the wall.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) "The youth are the backbone of India" explain.

Or

(b) Why real objectives in education are important? Explain with the views of Livingstone.

12. (a) Justify the title "On Habits".

Or

(b) Discuss the development of the poverty motif over the course of the novel "Crime and Punishment".

13. (a) Why did Margaret Atwood like to read animal stories?

Or

(b) Explain Sarojini Naidu's visions of Love, Religion and Patriotism.

14. (a) Fill in the blanks using present perfect tense :

(i) She ———————— (finish) her breakfast now.

(ii) He recently ———————— a new car.(buy)

(iii) I ———————— (see) people die of starvation.

(iv) India ———————— free since 1947.(be)

(v) They ———————— (live) here for many year.

Or

(b) What are the types of sentences? Explain it briefly.

15. (a) Identify the sentence pattern for the following sentences.

(i) My friend lent me ten rupees.

(ii) The sun has already set.

(iii) They were flying kites.

(iv) Bring me that book.

(v) Please get me two tickets.

Or

(b) Fill in the blanks with suitable prepositions.

(i) We congratulated her ———————— her success.

(ii) Don't waste your time ———————— useless things.

(iii) Thank you ———————— your kind offer.

(iv) I explained my difficulty ———————— her.

(v) He spends a lot of money ———————— clothes.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) What are the real objectives of Education?

Or

(b) Write the summary of the story "On the Power of Youth".

17. (a) Illustrate the views of Indira Gandhi on youth and betterment of India.

Or

(b) Show how the title 'Crime and Punishment' is a satire on the new-fangled notions of teaching that a teacher has to practise.

18. (a) Elucidate the uses of Present, Present Continuous, Present Perfect and Present Perfect Continuous tenses with examples.

Or

(b) Explain how Auxiliaries are used as Modals?

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

PROGRAMMING IN C

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are keywords? Give example.

2. Write the precedence of arithmetic operators.

3. How will you define dynamic arrays?

4. What is the use of '*break*' statement?

5. Write a program to get a character from the console input.

6. Define String. Give examples.

7. What is a pointer?

8. When will you use Union?

9. What is a file?

10. Write short note on : Command Line arguments.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the fundamental data types in C?

Or

(b) Write short note on formatted input and output statements.

12. (a) Write a C program to find the maximum of three given numbers.

Or

(b) Briefly explain about '*switch*' statement. Give example.

13. (a) Write a C program to check the given string is palindrome or not.

Or

(b) How will define and call the function? Give example.

14. (a) Distinguish between '*Structure*' and '*Union*'.

Or

(b) How will you declare and initialize pointers? Explain.

15. (a) Write a C program to open and close a file.

Or

(b) Write short note on : Random Access File.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Discuss about various types of operators in C with suitable examples.

Or

(b) Explain in detail about various looping statements with examples.

17. (a) Discuss about various types of string handling functions.

Or

(b) Write a program to sort N given numbers in Ascending order.

18. (a) Explain about 'structure' within 'structure'. Give example.

Or

(b) Write short note on : *malloc(), calloc() and realloc()* functions.

———————

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

ALGEBRA AND CALCULUS – Allied

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. If α, β, γ are the roots of $x^3 + px^2 + qx + \gamma = 0$, find $\sum \alpha^2 \beta$.

2. State Descarte's rule of sign.

3. Define singular matrix.

4. Find the eigen values of the matrix $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$.

5. Find $\frac{dy}{dx}$ if $y = x^2 + 2$.

6. Define convexity of a curve.

7. Evaluate $\int \frac{dx}{x^2 + 2x + 5}$.

8. Show that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$.

9. Solve : $e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$.

10. Solve : $(D^2 - 5D + 6)y = 0$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Solve the equation $81x^3 - 18x^2 - 36x + 8 = 0$ whose roots are in harmonic progression.

Or

(b) Increase by 7 the roots of the equation $3x^4 + 7x^3 - 15x^2 + x - 2 = 0$.

12. (a) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 8 & 2 \\ 4 & 9 & -1 \end{bmatrix}$.

Or

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

13. (a) Show that the minimum value of $x \log x$ is $-e^{-1}$.

Or

(b) Find the points of inflexion on the curve $y = (\log x)^3$.

14. (a) Evaluate : $\int \frac{1}{(x+1)(x+2)} dx$.

Or

(b) Obtain the half-range cosine series for $f(x) = x$ in $(0, \pi)$.

15. (a) Solve : $\frac{dy}{dx} = \frac{x+7y+2}{3x+5y+6}$.

Or

(b) Solve : $(D^2+16)y = e^{-3x}$.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Find the equation whose roots are the cubes of the roots of $x^4 - x^3 + 2x^2 + 3x + 1 = 0$.

Or

(b) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 3 & 0 \\ 5 & 6 & 7 \end{bmatrix}$.

17. (a) If $u = xy\, f\left(\frac{y}{x}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u$.

Or

(b) Evaluate $\int \frac{6x+5}{\sqrt{6+x-2x^2}}$.

18. (a) Determine the Fourier series of $f(x) = x^2$ in $0 < x < 2\pi$.

Or

(b) Solve : $(D^4 + D^2 + 1)y = x^2 + e^x \sin 2x$.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

இலக்கணமும் படைப்பிலக்கியமும்

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

பகுதி அ $(10 \times 2 = 20)$

எல்லா வினாக்களுக்கும் விடை தருக.

1. மெய்ம்மயக்கம் என்றால் என்ன?

2. சார்பெழுத்துக்கள் எவை? விவரி.

3. பாரதிதாசன் பற்றி குறிப்பு வரைக.

4. கண்ணதாசன் பற்றி நீவிர் அறிவன யாவை?

5. அசோகமித்ரனின் சிறப்புகள் யாவை?

6. புதுமைப்பித்தன் எத்துறையில் வல்லவர்?

7. இணைய இதழ்கள் பற்றி விவரி.

8. தமிழ் வளர்ச்சித் துறையின் பணிகள் யாவை?

9. புதுக்கவிதையின் இலக்கணம் தருக.

10. ‘கட்டுரை’ அமைப்பினை விவரி.

பகுதி ஆ (5 × 5 = 25)

எல்லா வினாக்களுக்கும் விடை தருக.

11. (அ) மொழிக்கு முதலில் வரும் எழுத்துக்களைச் சான்றுடன் தருக.

(அல்லது)

(ஆ) உடம்படுமெய்யினைச் சான்றுடன் விளக்குக.

12. (அ) பாரதியின் காணிநிலம் பாடலின் சிறப்பினை விவரி.

(அல்லது)

(ஆ) சாகாதவானம் கவிதை உணர்த்தும் செய்திகள் எவை?

13. (அ) தாய்ப்பசு கதைக்கருவை விவரி.

(அல்லது)

(ஆ) தந்தை மகனுக்கு ஆற்றும் நன்றி யாது?

14. (அ) படைப்பிலக்கிய வளர்ச்சியை ஆய்க.

(அல்லது)

(ஆ) இணையமும் தமிழும் பற்றி விளக்குக.

15. (அ) மரபுக் கவிதையின் அமைப்பினை விவரி.

(அல்லது)

(ஆ) சிறுகதையின் சிறப்பம்சங்கள் யாவை?

பகுதி இ (3 × 10 = 30)

எல்லா வினாக்களுக்கும் விடை தருக.

16. (அ) வல்லினம் மிகும், மிகாத இடங்களைச் சான்றுடன் விவரிக்க.

(அல்லது)

(ஆ) பாரதிதாசன் கால கவிதை வளர்ச்சியை விவரி.

17. (அ) புதுக்கவிதையின் தோற்றமும் வளர்ச்சியும் பற்றி எழுதுக.

(அல்லது)

(ஆ) இணையத்தின் இன்றைய வளர்ச்சியை விவரிக்க.

18. (அ) இணைய வேலைவாய்ப்புகளை ஆய்வு செய்க.

(அல்லது)

(ஆ) 'அன்பு' எனும் தலைப்பில் இருபது வரிகளில் கவிதை படைத்திடுக.

CP-7892

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21A

B.Sc. (Computer Science) DEGREE

EXAMINATION, NOVEMBER 2017

Second Semester

Part I

ARABIC – PROSE

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the meaning of الحجرات? Which verse contains this term - الحجرات in this chapter?

2. Write the English equivalents for the following :

(a) أجر (b) فاسق

(c) الإيمان (d) الأرض

3. Who bestowed wisdom on Luqman (A.S.)?

4. What do you know about

أمر بالمعروف ونهي عن المنكر

5. For whom جنّٰت النعيم is promised in سورة لقمان?

6. Write the English equivalents for the following :

(a) الحكمة

(b) ظلم عظيم

(c) الصدور

(d) سبعة أبحر

7. What is meant by بني الإ سلام على خمس?

8. Write the meaning of the following :

"من حسن إسلام المرء تركه ما لا يعنيه".

9. What is meant by "وخالق الناس بخلق حسن"?

10. Write the meaning for the following :

واللّٰه في عون العبد ماكان العبد في عون أخيه

Part B (5 × 5 = 25)

Answer **all** questions.

Explain the following :

11. (a) يَا أَيُّهَا الَّذِينَ آمَنُوا إِن جَاءَكُمْ فَاسِقٌ بِنَبَإٍ فَتَبَيَّنُوا أَن تُصِيبُوا قَوْمًا بِجَهَالَةٍ فَتُصْبِحُوا عَلَىٰ مَا فَعَلْتُمْ نَادِمِينَ.

Or

(b) قُلْ أَتُعَلِّمُونَ اللَّهَ بِدِينِكُمْ وَاللَّهُ يَعْلَمُ مَا فِي السَّمَاوَاتِ وَمَا فِي الْأَرْضِ ۚ وَاللَّهُ بِكُلِّ شَيْءٍ عَلِيمٌ.

12. (a) إِنَّ الَّذِينَ آمَنُوا وَعَمِلُوا الصَّالِحَاتِ لَهُمْ جَنَّاتُ النَّعِيمِ خَالِدِينَ فِيهَا ۖ وَعْدَ اللَّهِ حَقًّا ۚ وَهُوَ الْعَزِيزُ الْحَكِيمُ.

Or

(b) وَإِذْ قَالَ لُقْمَانُ لِابْنِهِ وَهُوَ يَعِظُهُ يَا بُنَيَّ لَا تُشْرِكْ بِاللَّهِ ۖ إِنَّ الشِّرْكَ لَظُلْمٌ عَظِيمٌ.

13. (a) وَوَصَّيْنَا الْإِنْسَانَ بِوَالِدَيْهِ حَمَلَتْهُ أُمُّهُ وَهْنًا عَلَىٰ وَهْنٍ وَفِصَالُهُ فِي عَامَيْنِ أَنِ اشْكُرْ لِي وَلِوَالِدَيْكَ إِلَيَّ الْمَصِيرُ.

Or

(b) وَلَئِن سَأَلْتَهُم مَّنْ خَلَقَ السَّمَاوَاتِ وَالْأَرْضَ لَيَقُولُنَّ اللَّهُ ۚ قُلِ الْحَمْدُ لِلَّهِ ۚ بَلْ أَكْثَرُهُمْ لَا يَعْلَمُونَ.

14. (a) قَالَ رَسُولُ اللّٰهِ (صَلَّى اللّٰه عَلَيْهِ وَسَلَّمَ) "دَعْ مَا يُرِيبُك إِلَى مَا لَا يُرِيبُك".

Or

(b) قَالَ رَسُولُ اللّٰهِ (صَلَّى اللّٰه عَلَيْهِ وَسَلَّمَ) "لَا يُؤْمِنُ أَحَدُكُمْ حَتَّى يُحِبَّ لِأَخِيهِ مَا يُحِبُّ لِنَفْسِهِ".

15. (a) قَالَ رَسُولُ اللّٰهِ (صَلَّى اللّٰه عَلَيْهِ وَسَلَّمَ) "إِنَّ مِمَّا أَدْرَكَ النَّاسُ مِنْ كَلَامِ النُّبُوَةِ الْأُولَى: إِذَا لَمْ تَسْتَحِ فَاصْنَعْ مَا شِئْت".

Or

(b) قَالَ رَسُولُ اللّٰهِ (صَلَّى اللّٰه عَلَيْهِ وَسَلَّمَ) "قُلْت : يَا رَسُولَ اللّٰه ! قُلْ لِي فِي الْإِسْلَامِ قَوْلًا لَا أَسْأَلُ عَنْهُ أَحَدًا غَيْرَك، قَالَ : قُلْ : آمَنْت بِاللّٰهِ ثُمَّ اسْتَقِمْ".

Part C (3 × 10 = 30)

Answer **all** questions.

Translate the following and explain in detail :

16. (a) إِنَّمَا الْمُؤْمِنُونَ الَّذِينَ آمَنُوا بِاللّٰهِ وَرَسُولِهِ ثُمَّ لَمْ يَرْتَابُوا وَجَاهَدُو بِأَمْوَالِهِمْ وَأَنفُسِهِمْ فِي سَبِيْلِ اللّٰهِ ۚ أُوْلٰئِكَ هُمُ الصَّادِقُونَ.

Or

(b) يَا بُنَيَّ إِنَّهَا إِن تَكُ مِثْقَالَ حَبَّةٍ مِّنْ خَرْدَلٍ فَتَكُن فِي صَخْرَةٍ أَوْ فِي السَّمَاوَاتِ أَوْ فِي الْأَرْضِ يَأْتِ بِهَا اللّٰهُ ۚ إِنَّ اللّٰهَ لَطِيفٌ خَبِيرٌ.

17. (a) اِنَّ اللّٰهَ عِنْدَهُ عِلْمُ السَّاعَةِ وَيُنَزِّلُ الْغَيْثَ وَيَعْلَمُ مَا فِي الْأَرْحَامِ ۖ وَمَا تَدْرِي نَفْسٌ مَّاذَا تَكْسِبُ غَدًا ۖ وَمَا تَدْرِي نَفْسٌ بِأَيِّ أَرْضٍ تَمُوتُ ۚ إِنَّ اللّٰهَ عَلِيمٌ خَبِيرٌ.

Or

(b) عَنْ أَمِيرِ الْمُؤْمِنِيْنَ أَبِي حَفْصٍ عُمَرَ بْنِ الْخَطَّابِ رَضِيَ اللّٰهُ عَنْهُ قَالَ : سَمِعْتَ رَسُولَ اللّٰهِ صلى اللّٰه عليه وسلم

يَقُولُ : ”إِنَّمَا الْأَعْمَالُ بِالنِّيَّاتِ، وَإِنَّمَا لِكُلِّ امْرِئٍ مَا نَوَى، فَمَنْ كَانَتْ هِجْرَتُهُ إِلَى اللّٰهِ وَرَسُولِهِ فَهِجْرَتُهُ إِلَى اللّٰهِ وَرَسُولِهِ، وَمَنْ كَانَتْ هِجْرَتُهُ لِدُنْيَا يُصِيبُهَا أَوْ امْرَأَةٍ يَنْكِحُهَا فَهِجْرَتُهُ إِلَى مَا هَاجَرَ إِلَيْهِ“.

18. (a) عَنْ أَبِي سَعِيدٍ الْخُدْرِيِّ رَضِيَ اللّٰهُ عَنْهُ قَالَ سَمِعْت رَسُولَ اللّٰهِ صلى اللّٰه عليه وسلم يَقُولُ : ”مَنْ رَأَى مِنْكُمْ مُنْكَرًا فَلْيُغَيِّرْهُ بِيَدِهِ، فَإِنْ لَمْ يَسْتَطِعْ فَبِلِسَانِهِ، فَإِنْ لَمْ يَسْتَطِعْ فَبِقَلْبِهِ، وَذٰلِكَ أَضْعَفُ الْإِيمَانِ“.

Or

(b) عَنْ النَّوَّاسِ بْنِ سَمْعَانَ رَضِيَ اللّٰـهُ عَنْهُ عَـنْ الـنَّـبِـيِّ صلـى اللّٰـه عـلـيـه وسـلـم قَـالَ : "الْـبِـرُّ حُـسْـنُ الْـخُـلُـقِ، وَالإِثْـمُ مَا حَاكَ فِي صَـدْرِك، وَكَرِهْت أَنْ يَطَّلِعَ عَـلَـيْـهِ الـنَّـاسُ".

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

ENGLISH – PROSE, EXTENSIVE READING AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Kalam is proud of having worked with three great men. Who are they?

2. Why does Mr. Bumble refuse to commission the painter?

3. What do sick people do in ancient times?

4. Who killed Gandhi?

5. How did Lynd become a nationalist?

6. What was the agreement between Bob and Jimmy Wells twenty years before, when they parted?

7. What did the conjuror do with the gold watch of the Quick Man?

8. Why did the astrologer tell his wife that a great load was off his mind?

9. What is pronoun?

10. What is superlative degree? Give example.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What is wrong with Indians, according to Kalam?

Or

(b) Why cannot the artist fit into a totalitarian set-up?

12. (a) How does Nehru blame himself?

Or

(b) What purpose can a referee serve at public debates, according to Lynd?

13. (a) How does Bob get arrested?

Or

(b) How does the astrologer tackle Guru Nayak?

14. (a) How did the old father test the skills learnt by his four sons?

Or

(b) What is Phrase and Simple sentence? Explain with examples.

15. (a) Jot down the changes while transforming the exclamatory sentences of Direct speech into Indirect speech.

Or

(b) Write a paragraph on the Formation of Comparative and Superlative'.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) What are the conditions needed to be fulfilled for realizing the vision, according to Kalam?

Or

(b) What are Robert Lynd's views on the efficacy or otherwise of arguing?

17. (a) Show how the conjuror takes revenge on the Quick Man.

Or

(b) Elucidate how naturalism and supernaturalism are blended in the short story *The Four Brothers*?

18. (a) What are the kinds of pronouns? Explain in detail.

Or

(b) What is subordinate clause? What are the kinds of subordinate clauses? Explain in detail.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

PROGRAMMING IN C++

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Name any three Object oriented Programming Languages.

2. What are the applications of Object Oriented Programming?

3. Define : Object.

4. What do you meant by Dynamic Binding?

5. What does '*this*' pointer point to?

6. What is virtual base class?

7. List different file modes.

8. What is file pointer?

9. List any two exceptions.

10. What is user defined templates?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the benefits of Object Oriented Programming?

Or

(b) Write short note on reference variables.

12. (a) Briefly explain about static member functions.

Or

(b) Write short note on : Copy Constructor.

13. (a) Write short note on : Abstract classes.

Or

(b) Briefly explain about unformatted I/O operations with example.

14. (a) Explain about error handling during file operations.

Or

(b) Write a C++ program to illustrate command line arguments.

15. (a) What is template? Explain about user defined templates.

Or

(b) Write a C++ program that demonstrates the concept of rethrowing an exception.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Discuss in detail about various control structures in C++. Give example.

Or

(b) Write a C++ program to illustrate dynamic initialization of objects.

17. (a) Explain multiple inheritance and multilevel inheritance with suitable example.

Or

(b) Write a C++ program to open and close a file using constructors.

18. (a) Distinguish between class template and function template with suitable example.

Or

(b) Write a C++ program that demonstrates the use of friend function with example.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Computer Science

NUMERICAL ANALYSIS AND STATISTICS — Allied

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find $\Delta(\log x)$.

2. Prove that $E = \left(\frac{\Delta}{\delta}\right)^2$.

3. What is the order of local error is trapezoidal rule?

4. What is the condition for Simpson's 3/8 rule?

5. State Adam's predictor corrector formula.

6. Solve $\frac{dy}{dx} = -y,\ y(0) = 1$ to find $y(0 \cdot 01)$ using Euler's method.

7. Find the mean of x_1, x_2, x_3, x_4, x_5.

8. If X and Y are two random variables such that $Y \leq X$, then prove that $E(Y) \leq E(X)$, provided all the expectations exist.

9. Define binomial distribution.

10. What is the moment generating function of normal distribution?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Find the smallest positive root of $x^3 - 2x + 0.5 = 0$ by Newton's Raphson method.

Or

(b) Evaluate $\Delta\left(\tan^{-1} x\right)$.

12. (a) Evaluate $\int_{-3}^{3} x^4 \, dx$ by using Simpson's rule. Verify your result of actual integration.

Or

(b) Dividing the range into 10 equal parts, find the approximate value of $\int_{0}^{\pi} \sin x \, dx$ by Trapezoidal rule.

13. (a) Using Taylor's series method compute $y(0.1)$ correct to 4 decimal places if $y(x)$ satisfies $y' = x + y,\ y(0) = 1$.

Or

(b) Given $y' = xy + y^2,\ y(0) = 1$. Compute $y(0.1)$ by fourth order Runge-Kutta method.

14. (a) Prove that for any discrete distribution, standard deviation is not less than mean deviation from mean.

Or

(b) Obtain the equation of two lines of regression from the following :

X :	65	66	67	67	68	69	70	72
Y :	67	68	65	68	72	72	69	71

15. (a) A and B play a game in which their chances of winning are in the ratio 3 : 2. Find A's chance of winning at least three games out of the five games played.

Or

(b) Find the median of normal distribution.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Find the real root of the equation $\sin x = 1 + x^2$ between $(-2, -1)$ to 3 decimal places by Newton's Raphson method.

Or

(b) Find a real root of the equation $x^3 + x^2 - 1 = 0$ by iteretion method.

17. (a) Solve the following system of equations, by Gauss-Jordan elimination method :

$$10x_1 + x_2 - x_3 = 11.19$$
$$x_1 + 10x_2 + x_3 = 20.08 \quad .$$
$$-x_1 + x_2 + 10x_3 = 35.61$$

Or

(b) Using Adam's method find $y(0.4)$ given that

$$y' = \frac{xy}{2},\ y(0.1) = 1.01,\ y(0.2) = 1.022,\ y(0.3) = 1.023\ .$$

18. (a) A coin is tossed until a head appears. What is the expectation of the number of tosses required?

Or

(b) If X and Y are independent poisson variate with means T_1 and T_2 respectively, find the probability that (i) $X + Y = K$ and (ii) $X = Y$.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

ENVIRONMENTAL STUDIES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short answer to the following :

1. Deforestation

2. Conflicts

3. Salinity

4. Desertification

5. Degradation

6. Food Chain

7. Biodiversity

8. *In situ*

9. Algal bloom

10. Bioaccumulation

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Why should create a public awareness on conservation of Nature? explain.

Or

(b) Write a note on renewable and non renewable energy resources.

12. (a) Write the conflicts over water resources.

Or

(b) Explain the role of an individual in conservation of natural resources.

13. (a) Give an account on structural components of an ecosystem.

Or

(b) What are the characteristic features of a simple ecosystem? explain.

14. (a) Write the method of assessing the value of Biological Resources.

Or

(b) Explain the causes of water pollution.

15. (a) List out the impact of thermal pollution.

Or

(b) Write an essay on the importance of filed visit and observation of nature.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Discuss the importance of knowledge on environment to the public.

Or

(b) Explain the role of individual in conservation of natural resources.

17. (a) Give an account on induced landslide and soil erosion.

Or

(b) Explain the concept of an ecosystem.

18. (a) Give a detailed account on energy flow in the ecosystem.

Or

(b) Describe the major threats to Biodiversity.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

COMMUNICATION SKILLS – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is communication?

2. What are the barriers of communication?

3. Frame questions using subject, verb and adverb.

4. Define Clause.

5. Use appropriate form of prepositions :

   (a) Solomon was born ———— Monday.

   (b) English is spoken all ———— the world.

6. Define Infinitive with example.

7. What is rising intonation?

8. Define Secondary stress with example.

9. What are SQ3R?

10. What is Skimming for information?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the objectives of communication?

Or

(b) How can barriers be reduced in communication?

12. (a) Give a brief account of Relative clauses.

Or

(b) What are the rules for changing simple sentence into compound sentence?

13. (a) Explain the adjectives that often get confused.

Or

(b) Write about zero articles.

14. (a) Explain Vowel and Consonants with examples.

Or

(b) Write the importance of primary stress in the sentences.

15. (a) Why is it so important to know how to read?

Or

(b) Differentiate skimming and scanning.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) What are the ways of challenging the barriers of communication?

Or

(b) How will you transform Imperative sentences from Active voice into Passive voice? Explain with examples.

17. (a) Write an essay on three models of conditional clauses.

Or

(b) What are the characteristics of Good Handwriting?

18. (a) What will you eliminate prepositions in unnecessary places? Give examples.

Or

(b) Write briefly the Mechanics of Handwriting.

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CP–7898

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

DIGITAL COMPUTER FUNDAMENTALS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Convert the following : $(0.6875)_{10} = (?)_2$.

2. Find the 10's complement of $(52520)_{10}$.

3. State Demorgan's Theorem.

4. What are universal gates?

5. Write short note on : Master slave flip flops.

6. Define Sequential logic circuit.

7. Draw the logic diagram of Half subtractor.

8. Define parallel binary adder.

9. Write the two types of data representations.

10. Define Gray code.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Convert hexadecimal number 2AC5.D to decimal.

Or

(b) Using 9's complement subtract $(72532)_{10}$ — $(3250)_{10}$.

12. (a) Obtain the simplified expression in sum of product form $f(x, y, z) = \sum(0, 2, 4, 5, 6)$.

Or

(b) Implement the Boolean function AB' + CD USING logic gates.

13. (a) Explain the operation of 4:1 multiplexer with neat diagram.

Or

(b) Explain the 4-bit binary counter with neat diagram.

14. (a) Draw the logic diagram of full adder and explain with its truth table.

Or

(b) Explain the 4-bit parallel binary subtractor.

15. (a) Discuss about floating point data representation format.

Or

(b) What is error? Explain the need for error detection codes.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Discuss in detail about alpha numeric and character codes.

Or

(b) Simplify the following Boolean function using Quine-Mccluskey method. $F(w, x, y, z) = \sum(0, 1, 2, 8, 10, 11, 14, 15)$.

17. (a) Explain the serial-in serial-out and serial-in and parallel-out shift registers.

Or

(b) Discuss in detail about the construction of ALU.

18. (a) Explain about various types of error detection codes.

Or

(b) Draw the logic diagram of BCD counter and explain it with state table.

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CP–7899

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Computer Science

JAVA PROGRAMMING

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Java Virtual Machine?

2. What are the applications of Object Oriented Programming?

3. List the special operators in Java.

4. List any two mathematical functions in Java.

5. What are final variables?

6. List any two Wrapper classes.

7. What are java naming conventions?

8. What are the types of errors?

9. How will you run applet?

10. Define Applet.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Briefly explain the features of java?

Or

(b) What is type casting? Give examples.

12. (a) Write short note on: Type conversion.

Or

(b) Briefly explain about nested if structure with example.

13. (a) Write short note on: overriding methods.

Or

(b) Distinguish between class and interface.

14. (a) Define thread. Explain the life cycle of a thread.

Or

(b) Discuss the use of *'finally'* statement in handling exception.

15. (a) Explain the life cycle of an applet.

Or

(b) Write an applet program to draw circle.

Part C

(3 × 10 = 30)

Answer **all** questions.

16. (a) Discuss in detail about the basic concepts of object oriented programming.

Or

(b) Explain various type of operators used in Java.

17. (a) How will you define, implement and access interface? Explain with example

Or

(b) How will you create, access and use package? Explain with example.

18. (a) Explain in detail about multithreaded programming with example program.

Or

(b) Write an applet program for the following;

(i) Getting input from the user. (5)

(ii) Draw rectangle. (5)

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CP–7900

Sub. Code
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B.Sc. (Computer Science) DEGREE EXAMINATION, NOVEMBER 2017

DATA STRUCTURE AND ALGORITHMS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define data structure.

2. What is array?

3. What are the applications of stack?

4. Convert the following infix notation to postfix.

   (A + B) * (C – D) / E

5. Draw various types of binary tree.

6. Define Hashing.

7. What is binary search technique?

8. What is the time and space complexity of quick sort?

9. What is the use of Big oh notation.

10. Write short note on: Pseudo code for expressing algorithms.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short note on : Merging Lists.

Or

(b) Explain the array representation of lists.

12. (a) Write an algorithm to find the factorial of N using recursion.

Or

(b) How will you represent stack? Explain.

13. (a) How will you perform insert and delete operations in binary search tree? Explain.

Or

(b) Explain the in-order binary tree traversal method with diagram.

14. (a) Briefly explain about linear search technique with suitable example.

Or

(b) Discuss the time and space complexity of binary search algorithm.

15. (a) Write short note on: Omega Notation.

Or

(b) Write short note on: Theta Notation.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) What is traversal of list? How will you search and retrieve an element from the list?

Or

(b) Explain the operations of queue with example. What are its applications?

17. (a) Discuss in detail about finding shortest path using *Dijkstra's* Algorithm.

Or

(b) Explain in detail about selection sorting technique with example.

18. (a) Discuss about the time and space complexity of algorithms with suitable example.

Or

(b) How will you insert and delete elements from doubly linked list? Explain.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

APPLIED PHYSICS – I (Allied)

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define electric intensity. Give its unit.

2. What is a capacitor?

3. Define susceptibility.

4. What is hysteresis?

5. State Laplace law.

6. Write down the Fleming's left hand rule.

7. State the laws of electromagnetic induction.

8. What is meant by eddy current?

9. Define power factor in an AC circuit.

10. What is selectivity?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Obtain an expression for the intensity due to a charged sphere.

Or

(b) Derive expressions for equivalent capacity, when two capacitors are connected in (i) series (ii) parallel.

12. (a) Deduce the relation between permeability and susceptibility.

Or

(b) What is a magnetic shell? Obtain its potential at any point.

13. (a) Derive an expression for the intensity at a point due to a straight conductor carrying current.

Or

(b) Explain the calibration of law range voltmeter.

14. (a) What is a transformer? Give its theory.

Or

(b) Establish the relation between induced emf and mutual inductance.

15. (a) Obtain an expression for the impedance of an AC circuit containing resistance and inductance in series.

Or

(b) What is a choke? Explain its function and user.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) State and prove Gauss theorem in electrostatin.

Or

(b) Discuss the method of drawing hysteresis curve by magnetometer method.

17. (a) Describe the construction and theory of carey-Foster's bridge.

Or

(b) Draw the circuit of Anderson's bridge and describe the method of determining self inductance using it.

18. (a) Give the theory of series resonance circuit.

Or

(b) Write a notes:

(i) Equipotential surface. (3)

(ii) Conversion of galvanometer in to ammeter. (4)

(iii) Theory of transformer. (3)

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